

1. $\Sigma^n = (\text{Gen}, \text{Enc}, \text{Dec})$ // perfectly secure one-time pad over $\mathbb{Z}_n$
 $M = K = C = \mathbb{Z}_n$
 $\text{Enc}(K, m) = (m + K) \bmod n$ ($n = \text{any + integer}$)

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$$\Sigma^* = (\text{Gen}^*, \text{Enc}^*, \text{Dec}^*) \text{ for } M^* = A \times A \text{ where } A \rightarrow \{0, \dots, 25\}$$

$$M^* = A \times A, |M^*| = 26 \times 26 = 676 \text{ // msg. space is 676 } (n = 676)$$

$$x = (m_1) + 1(m_2) \cdot 26$$

↑ bijectively maps (m_1, m_2) to a unique integer from 0-675.

$\therefore D$ is a perfectly secure encryption scheme (each message pair has a unique mapping)

2. $M^* = \{0, 1, \dots, 99\} \cup \{\text{NULL}, \text{ERROR}, \text{EOF}\}$ $\Sigma^n = (\text{Gen}, \text{Enc}, \text{Dec})$
 ↳ perfectly secure over $\mathbb{Z}$
 Construct enc. scheme $\Sigma^*(\text{KeyGen}^*, \text{Enc}^*, \text{Dec}^*)$
 that is perfectly secure for $m \in M^*$, with encoding funct. $F: M^* \rightarrow \mathbb{Z}_n$

a. $n \geq 103$

↳ n must have at least as many elements as M^* ,
 M^* contains 100 integers and 3 symbols, total 103 msg.

b. $F(m) = m, m \in \{0, \dots, 99\}$

↳ control symbols can be mapped w/ integers (100, 101, 103)
 (still are distinct msgs., preserves 103 total)

c. Encryption alg. defined as $c = \Sigma^n \text{Enc}(K, F(m))$
 ?

Consider a specific instantiation where we select $n = 200$ for the underlying scheme, the mapping function F remains unchanged (use first 103 of 200 indices)

a. with $n = 200$ ($|K| = 200, |M^*| = 103$)

The scheme does satisfy perfect security

↳ there exists enough keys to uniquely match all possible messages

b. The fundamental reason for above is that, for any
 $m \in M^*, c = (F(m) + K) \bmod 200$ is unique?

↳ messages from 1...103 will all be uniquely represented

⋮

Dropdown questions 1-4

3. consider the scheme $\Sigma^n = (\text{Gen}, \text{Enc}, \text{Dec})$ defined over the vectorspace
 $\mathbb{Z}_n \times \mathbb{Z}_n$ with
 $M = C = K = \mathbb{Z}_n \times \mathbb{Z}_n$
 Gen : Sample a uniform pair $(K_1, K_2) \leftarrow \mathbb{Z}_n \times \mathbb{Z}_n$
 Enc : Output $c = (m_1 + K_1 \bmod n, m_2 + K_2 \bmod n)$

↳ Is the scheme described perfectly secure?

↳ Yes, by C, every message is encrypted by a unique set of keys

4. consider the encryption scheme $\Sigma^n = (\text{Gen}, \text{Enc}, \text{Dec})$ over $\mathbb{Z}_n$:

- $\mathcal{M} = \mathcal{C} = \mathbb{Z}_n$, $\mathcal{K} = \mathbb{Z}_n \times \mathbb{Z}_n$
- Gen: Sample a uniform pair $(k_1, k_2) \leftarrow \mathbb{Z}_n \times \mathbb{Z}_n$
- Enc: Output $c = m \cdot k_1 + k_2 \pmod{n}$

↳ Is scheme perfectly secure? No by C

↳ the linear coefficient of k_1 allows k_2 to be revealed

5. Consider the encryption scheme $\Sigma = (\text{Gen}, \text{Enc}, \text{Dec})$ with $\text{Enc}(k, m) = m + k \pmod{10}$

$$\mathcal{M} = \mathbb{Z}_{10}$$

$$\mathcal{C} = \mathbb{Z}_{10}$$

K sampling is non-uniform (follows table prob. dist.)

// Distinguishing game

$$\text{Adv}(A) = |\Pr[A(c) = 1 \mid b = 1] - \Pr[A(c) = 1 \mid b = 0]|$$

-if Challenger flips $1 \rightarrow c = 0 + k \pmod{10}$
 $0 \rightarrow c = 5 + k$

Alice: Outputs 1 if c is odd, 0 if even:

W1 $\hookrightarrow c$ is odd if $k \in \{1, 3, 5, 7, 9\}$

$$\Pr(\text{odd} \mid b=0) = \sum_{k=\text{odd}} = 0.2 + (0.05)4$$

$$0.2 + 0.2 = 0.4$$

W2 $\hookrightarrow c$ is odd if $k \in \{0, 2, 4, 6, 8\}$

$$\Pr(\text{odd} \mid b=1) = 0.4 + (0.05)4 = 0.6$$

$$\text{Adv}(A) = 0.6 - 0.4 = 0.2$$

Bob: Output 1 if $c \in \{5, 6, 7\}$

World 1: $c = 5, 6, 7$ if $k = 5, 6, 7$

$$\Pr = (0.05)3 = 0.15$$

World 2: $c = 5, 6, 7$ if $k = 0, 1, 2$

$$\Pr = 0.4 + 0.2 + 0.05 = 0.65$$

$$\text{Adv}(B) = 0.65 - 0.15 = 0.5$$

Eve: Receives $c=6$, what is the probability she is in World 1? $\rightarrow 0.8$

$$\Pr[\text{World 1} \mid c=6]$$

$$\Pr[m_b = 5 \mid c=6]$$

$$\Pr[b=1 \mid c=6] = \frac{\Pr[c=6 \mid b=1] \Pr(b=1)}{\Pr(c=6)} = \frac{(0.2)(0.5)}{0.125} = 0.8$$

$$\left. \begin{array}{l} G=5+k \\ k=1 \\ p=0.2 \\ G=0+k \\ k=6 \end{array} \right\} \Pr[c=6] = \frac{0.5(0.2+0.05)}{0.5(0.25)} = 0.125$$

6 / Bonus Problem :

Perfect security in a Two Time Setting :

$$m_0, m_1, m'_0, m'_1 \in \mathcal{M} \quad c_0, c_1 \in \mathcal{C}$$

$$\Pr[\text{Enc}(K, m_0) = c_0 \text{ and } \text{Enc}(K, m_1) = c_1] = \\ \Pr[\text{Enc}(K, m'_0) = c_0 \text{ and } \text{Enc}(K, m'_1) = c_1]$$

1. show that no deterministic encryption scheme w/ perfect correctness can satisfy :

Assume (for contradiction), there exists a deterministic encryption scheme w/ perfect correctness that satisfies

$\text{Enc}(K, m)$ is a fixed value for all (K, m)

pick any two distinct messages $m_0 \neq m_1 \in \mathcal{M}$, for key K :

$$c_0 := \text{Enc}(K, m_0), \quad c_1 := \text{Enc}(K, m_1)$$

↳ because of perfect correctness: $c_0 \neq c_1$

consider pairs:
 A: (m_0, m_0)
 B: (m_0, m_1)

$$\hookrightarrow \Pr[\text{Enc}(K, m_0) = c_0 \text{ and } \text{Enc}(K, m_1) = c_1] = 0 \text{ (since } c_0 \neq c_1)$$

but definition for perfect correctness implies

and $\text{Dec}(K, \text{Enc}(K, m)) = m$ with prob. 1

$$\Pr[\text{Enc}(K, m_0) = c_0 \text{ and } \text{Enc}(K, m_1) = c_1] = 1$$

∴ we have a contradiction and the scheme does not satisfy perfect correctness

2. show that no randomized encryption scheme with perfect correctness can satisfy

- by definition of randomized encryption scheme, ciphertext results of encryption are different each time, even with the same key used.

- For perfect correctness, the same ciphertext should be produced using the same message and key, but this conflicts w/ randomization

∴ two-time encryption scheme cannot also be randomized